

## Multiple Choice

1. **Answer:** D

*Options:* A) total revenues from production are equal to implicit costs.; B) total revenues from production are equal to explicit costs.; C) explicit costs are higher than implicit costs.; D) total revenues from production are equal to the sum of explicit and implicit costs.; E) total revenues from production are lower than the sum of explicit and implicit costs.

*Note:* The correct answer is based on the concept of normal profit, which occurs when total revenues equal the total of both explicit and implicit costs, indicating that the firm is covering all its opportunity costs without generating economic profit.

2. **Answer:** A

*Options:* A) An increase in printing costs; B) Expectations of future surpluses; C) A decrease in the demand for a substitute in production; D) An increase in the literacy rate; E) A decrease in the cost of labor for bookbinding

*Note:* An increase in printing costs raises the overall production expenses for publishers, leading to a reduction in the quantity of textbooks supplied at every price level, thus shifting the supply curve to the left.

3. **Answer:** C

*Options:* A) The debt is the sum of all past and future deficits.; B) When the debt is positive the deficit decreases.; C) When the deficit is negative the debt decreases.; D) The deficit is the accumulation of past debts.; E) When the debt is negative the deficit decreases.

*Note:* A negative deficit indicates a budget surplus, meaning the government is generating more revenue than it spends, which allows it to pay down existing debt, thereby decreasing the overall federal debt.

4. **Answer:** A

*Options:* A) Society does not get enough output.; B) There is perfect competition in the market, leading to an efficient outcome.; C) Monopolistically competitive firms produce at the minimum of the average cost curve.; D) There is a socially efficient allocation of resources.; E) The firm's profits are minimized, leading to an equitable distribution of resources.

*Note:* In a monopolistically competitive market, when marginal cost is below price, it indicates that the firm is producing less than the socially optimal output level, leading to a deadweight loss and underutilization of resources, resulting in societal inefficiency.

5. **Answer:** A

*Options:* A)  $TFC = AVC$  at all levels of output.; B)  $AVC + AFC = TC$ .; C)  $TFC = TC$  at all levels of output.; D)  $MC = AVC + AFC$ .; E)  $AFC = TC - AVC$ .

*Note:* The correct answer is based on the understanding that Total Fixed Costs (TFC)

do not change with the level of output, while Average Variable Costs (AVC) represent variable costs per unit of output, meaning that TFC and AVC are not equal at all levels of output; rather, TFC remains constant while AVC varies with output.

## True / False

6. **Answer:** False

*Options:* A) True; B) False

*Note:* A progressive income tax is designed to tax higher income levels at greater rates, but it does not cap personal wealth; rather, it allows individuals to accumulate wealth while contributing a fair share of their income to public resources.

7. **Answer:** False

*Options:* A) True; B) False

*Note:* An increase in private sector investment typically leads to higher interest rates as the demand for capital increases, which in turn causes Treasury bond prices to fall due to the inverse relationship between bond prices and interest rates.

8. **Answer:** False

*Options:* A) True; B) False

*Note:* The statement is false because the total change in aggregate demand resulting from an autonomous investment is calculated using the formula for the multiplier, which in this case would be  $1/(1 - MPC) = 10$ , leading to a total increase in aggregate demand of 1,000, *not* 500.

9. **Answer:** False

*Options:* A) True; B) False

*Note:* The statement is false because the money multiplier effect is based on the reserve requirement ratio, and a sale of government securities by the FED actually decreases the money supply, rather than increasing it.

10. **Answer:** False

*Options:* A) True; B) False

*Note:* U.S. coins are minted by the U.S. Department of the Treasury through the U.S. Mint, a government agency, which classifies them as fiat money rather than token money, as they are issued and regulated by a central authority.

## Long Form Response

11. **Answer:** The fundamental differences between the short-run and long-run Phillips curves lie in their shapes and the implications for inflation and unemployment within economic policy. The short-run Phillips curve is downward sloping, indicating an inverse relationship between inflation and unemployment; as inflation rises, unemployment tends to decrease, primarily due to nominal wage rigidity and price stickiness. Conversely, the long-run Phillips curve is vertical, suggesting that in the long term,

there is no trade-off between inflation and unemployment; the economy tends to gravitate towards the natural rate of unemployment regardless of the inflation rate. This distinction implies that while policymakers can exploit the short-run trade-off through expansionary policies to reduce unemployment, such measures may lead to accelerating inflation without permanently lowering unemployment in the long run. Therefore, effective economic policy must consider these differing dynamics to avoid the pitfalls of inflationary spirals.

*Note:* correct because it accurately describes the short-run Phillips curve's downward slope indicating a trade-off between inflation and unemployment due to nominal wage rigidity, while also correctly identifying the long-run Phillips curve's vertical position, which implies that this trade-off does not persist in the long term, thus influencing economic policy decisions.

12. **Answer:** An increase in the required reserve ratio mandates that banks hold a larger fraction of their deposits as reserves, thereby reducing the amount of funds available for lending. This contraction in lending capacity leads to a decrease in the money supply, as fewer loans translate into less money circulating in the economy. The broader implications include potential slowing of economic growth, as businesses and consumers face tighter credit conditions, which can stifle investment and consumption. Furthermore, higher reserve requirements can impact interest rates; as banks have less to lend, they may raise rates to allocate their limited funds, further dampening economic activity. Ultimately, such a policy could lead to reduced liquidity in the financial system, affecting overall economic stability.

*Note:* correct because it clearly articulates how an increase in the required reserve ratio restricts banks' lending abilities, leading to a contraction in the money supply and subsequently affecting economic growth through tighter credit conditions for both businesses and consumers.